

# **IBM University Engagement Project Submission**

*“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”*

# Project Title – AI Legal Aid Multi-Agent System

## Domain of Project – Legal Technology

| Student Name         | Recent Passport Photo |                                                                                    | Email ID             | Mobile number [WhatsApp] |
|----------------------|-----------------------|------------------------------------------------------------------------------------|----------------------|--------------------------|
| Mohammed Umeir Patel |                       |  | umeirp3636@gmail.com | 9373597210               |

# Problem Statement - AI Legal Aid Multi-Agent System

## The Challenge

The proposed solution for the **AI Legal Aid Multi-Agent System** is to design an intelligent, interactive platform that simplifies access to legal information, document analysis, and compliance guidance for individuals and small businesses. The assistant will integrate, **IBM Orchestrate, and IBM Models** to create a unified legal support system. **Knowledge retrieval** will be handled through Langflow pipelines, which ingest and semantically search legal documents, statutes, regulations, contracts, and case law to provide accurate and context-aware information. **Legal concept explanation** will be powered by IBM Models, delivering responses in simple, easy-to-understand language while also providing detailed legal insights for advanced users, making legal knowledge accessible to everyone. **Document analysis and specialized legal support** will be coordinated through IBM Orchestrate, enabling multiple AI agents to review contracts, identify risks, verify compliance with applicable regulations, conduct case research, and answer user-specific legal queries. Finally, a **conversational interface** will allow users to interact naturally with the assistant by asking legal questions or uploading legal documents, receiving structured summaries, highlighted clauses, compliance recommendations, and relevant legal references. This solution will reduce the time and effort required to understand legal information, improve awareness of legal rights and obligations, and provide practical legal guidance while clearly informing users that the system is intended for informational purposes only and is not a substitute for professional legal advice.

# Proposed Solution -

## Proposed Solution: AI Legal Aid Multi-Agent System

The proposed solution for the **AI Legal Aid Multi-Agent System** is to design an intelligent, interactive platform that simplifies access to legal information, document analysis, and compliance guidance for individuals and small businesses. The assistant will integrate , **IBM Orchestrate**, and **IBM Models** to create a unified legal support platform. **Legal knowledge retrieval** will be handled by Langflow pipelines, which ingest and semantically search legal statutes, regulations, contracts, policies, and case law to provide accurate, context-aware information. **Legal concept explanation** will be powered by IBM Models, delivering responses that range from beginner-friendly summaries to detailed legal interpretations, making complex legal concepts easy to understand for users with varying levels of legal knowledge. **Document analysis and workflow automation** will be managed through IBM Orchestrate, enabling specialized AI agents to review contracts, identify risky clauses, verify regulatory compliance, answer user-specific legal questions using Retrieval-Augmented Generation (RAG), and research relevant case precedents. Finally, a **conversational interface** will allow users to interact naturally with the assistant by asking legal questions or uploading legal documents, receiving structured summaries, highlighted clauses, compliance recommendations, and relevant legal references through an intuitive dashboard. This solution will reduce the time and cost associated with obtaining basic legal guidance, improve understanding of legal rights and obligations, assist users in identifying potential legal risks, and make legal information more accessible while clearly stating that the assistant provides informational support and is not a substitute for professional legal advice.

# Technology Used

- **IBM Orchestrate** -
  - Automates guided troubleshooting flows.
  - Provides diagnostic support for satellite communication issues.
  - Helps users follow structured problem-solving steps.
- **IBM Models**
  - Powers natural language understanding.
  - Generates simplified explanations of complex concepts (modulation, frequency bands, link budgets).
  - Offers layered responses for both beginners and advanced learners.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **RAG**– RAG to fetch real data first, then generate intelligent answers, making the satellite communication assistant Agent more trustworthy and effective.

# Orchestrate component Used -

1. **1) Chat Input** – Interface where users enter their queries about satellite communication (text/voice).
2. **2) IBM watsonx AI Agent** - Processes user input using Granite models to generate satellite communication and recommendations.
3. **3) Name of IBM Granite model** - Granite-4.0-8B-Instruct → Fast, efficient for real-time satellite communication recommendations
4. **Document Loader** – Loads textbooks, research papers, and technical documentation into the system for knowledge ingestion.
5. **LLM Chain** – Connects the language model with retrieved knowledge to generate simplified explanations of complex concepts.
6. **Conversational Chain** – Maintains context across multiple queries, enabling interactive and session-based responses.
7. **Prompt Template** – Structures user queries for consistent processing and ensures domain-specific responses.
8. **Output Parser** – Formats AI-generated responses into user-friendly text, diagrams, or troubleshooting steps.

# Agent Instructions Screenshot-

IBM watsonx Orchestrate

English

Contact Sales

Buy Now

29 trial days left

UP

Home / Manage agents / LexiGuard AI

Test agent

Deploy

LexiGuard AI

Model : GPT-OSS 120B — Open...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Behavior

Instructions

information through RAG, identifies important clauses, risks, and compliance issues, and then provides a clear, easy-to-understand summary with recommendations. It presents the results through an interactive dashboard while reminding users that the information is for guidance only and not a substitute for professional legal advice.

Specify what your agent should do, how it should respond and any restrictions it should follow.

Guidelines

Add Guideline +

Preview

Running in debug mode

9:46 PM

Hello, welcome to LexiGuard AI by watsonx Orchestrate

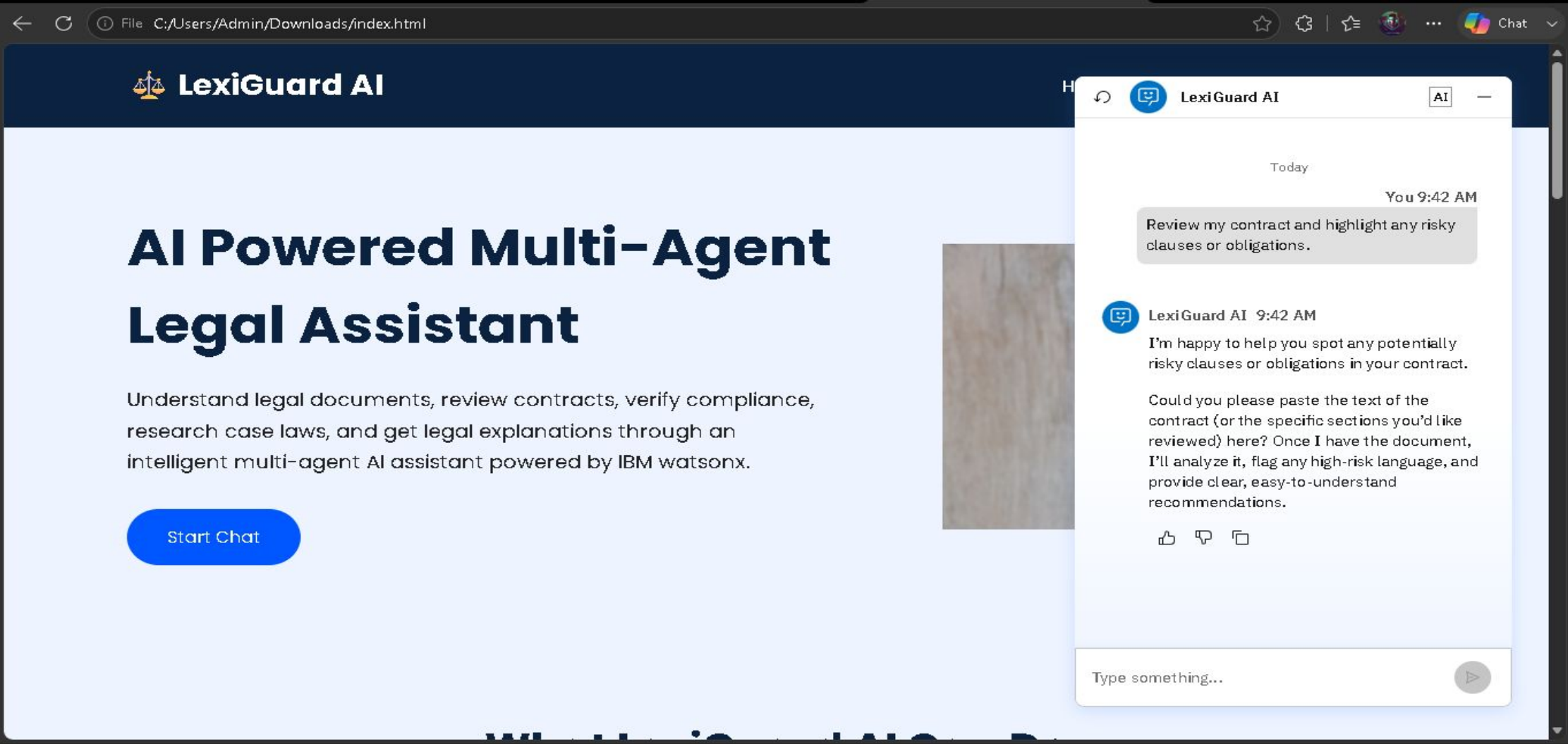
Accuracy of generated answers may vary. Please double-check responses.

Review my contract and highlight any risky clauses or obligations.

Explain my legal rights regarding this issue in simple language.

Type something...

Prompt with output Screenshot:





# Prompt with output Screenshot:

IBM watsonx Orchestrate

EnglishContact SalesBuy Now29 trial days left

Home / Manage agents / LexiGuard AI

Test agentDeploy

LexiGuard AI

Model : GPT-OSS 120B — Ope...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Profile

Describe what your agent does so people and other agents know when to use it.

Description\*

LexiGuard AI = receives a user's legal question or document, analyzes it using specialized AI agents, retrieves relevant laws and case information through RAG, identifies important clauses, risks, and compliance issues, and then provides a clear, easy-to-understand summary with recommendations. It presents the results through an interactive dashboard while reminding users that the information is for guidance only and not a substitute for professional legal advice.

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

What makes a good description?

Welcome message

Customize the welcome message from the agent on the home screen.

View all

Preview

Running in debug mode

LexiGuard AI 9:48 PM

I'm happy to help you spot any potentially risky clauses or obligations in your contract.

First, please upload the contract (or paste the text) you'd like reviewed. Once I have the document, I'll:

1. Parse the text and identify the key sections.

2. Highlight clauses that commonly pose risks (e.g., limitation of liability, indemnification, termination, confidentiality, non-compete, payment terms, change-of-law, force-majeure, etc.).

3. Summarize the main obligations for each party and flag any language that may be ambiguous, overly broad, or unfavorable.

4. Provide plain-English explanations and practical recommendations for how you might address or negotiate those points.

Please note: the analysis I provide is for informational purposes only and does not constitute legal advice. For definitive guidance, you should consult a qualified attorney who can consider the full context of your situation.

Whenever you're ready, upload the contract and I'll get started!

Type something...

# Prompt with output Screenshot:

IBM watsonx Orchestrate

EnglishContact SalesBuy Now

29 trial days left

UP

[Home](#) / [Manage agents](#) / LexiGuard AI

LexiGuard AI

Model : GPT-OSS 120B — Open...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Knowledge

Provide the specific knowledge your agent needs to accomplish its goal.

Knowledge sources

Add an existing knowledge source, upload files, or connect to content repositories to add knowledge to your agent.

Replace source +

legal data

by IBMid-691001STRB

legal document

Used by 1 agent

9:33 AM

Hello, welcome to LexiGuard AI by watsonx Orchestrate

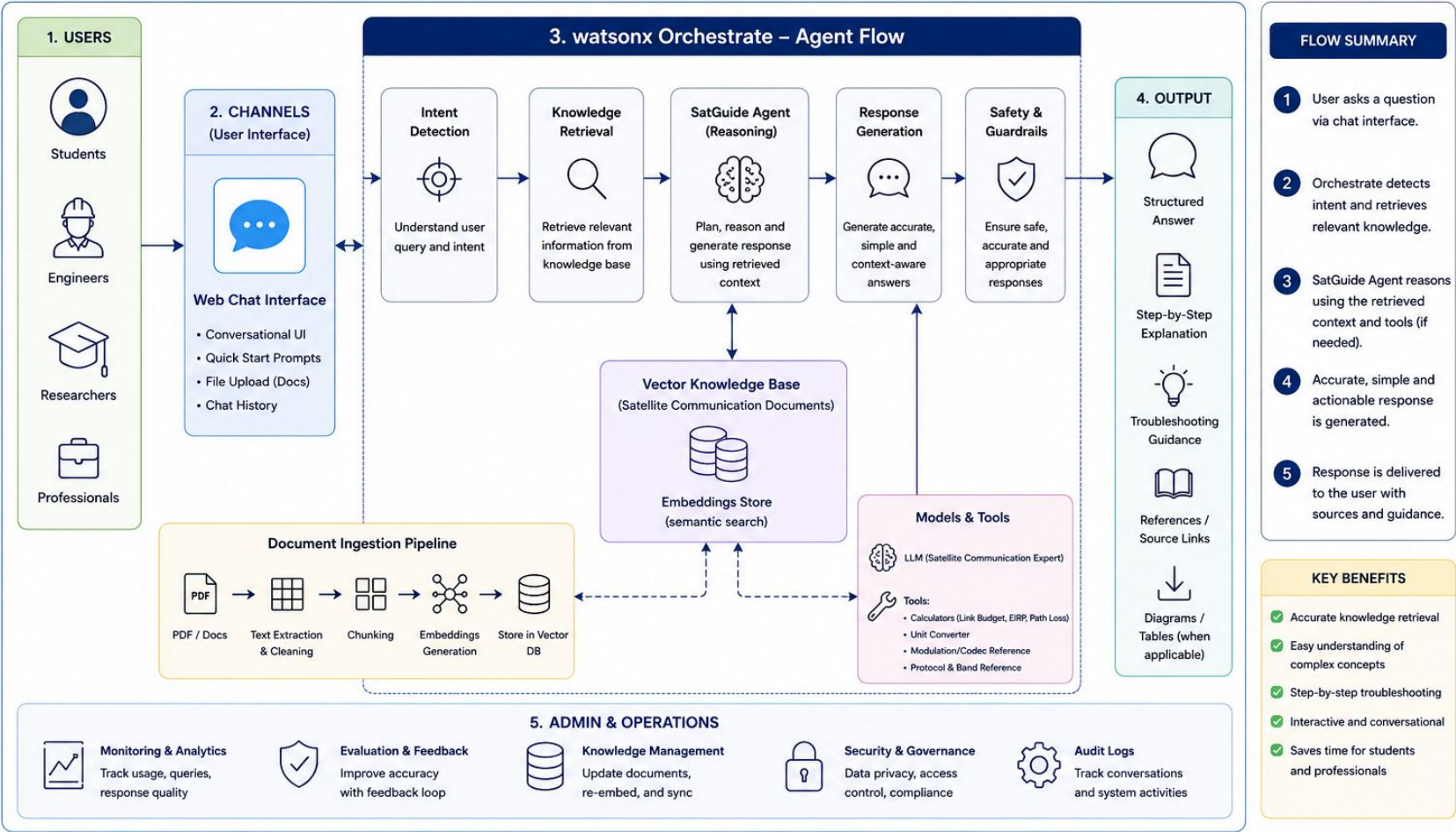
Accuracy of generated answers may vary. Use double-check responses.

Type something...

# Orchestrate architecture blueprint

## SatGuide – Smart Satellite Communication Knowledge Assistant

watsonx Orchestrate Architecture Blueprint



SatGuide helps users learn, understand and troubleshoot satellite communication systems intelligently.

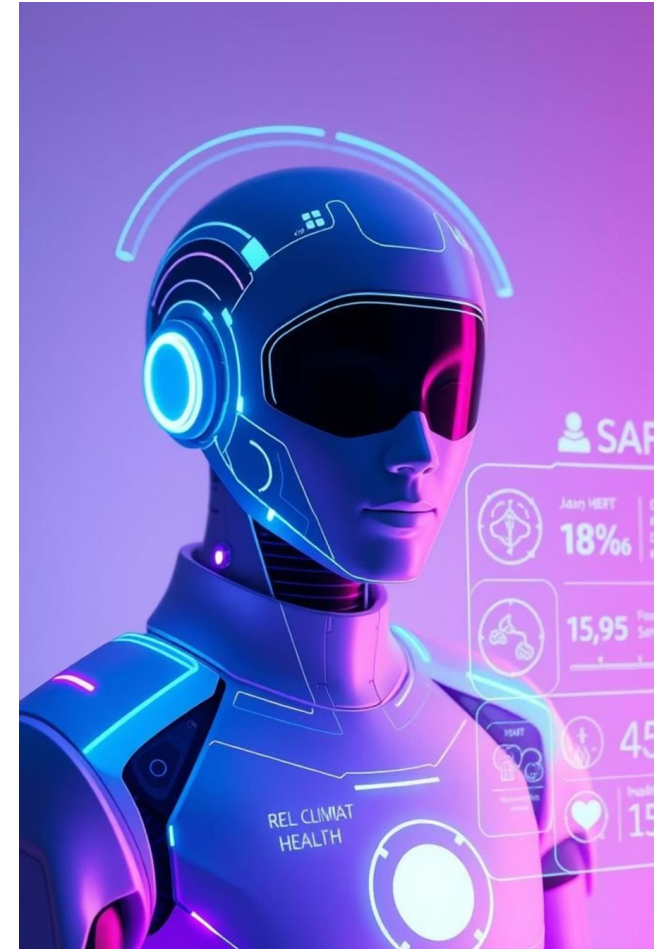


# Role of Agentic AI in the solution

## Role of Agentic AI in AI Legal Aid Multi-Agent System

The role of **Agentic AI** in the **AI Legal Aid Multi-Agent System** is to make the platform proactive, intelligent, and collaborative rather than simply responding to user queries. It autonomously retrieves relevant legal information, coordinates specialized AI agents, analyzes legal documents, and delivers context-aware guidance while maintaining the flow of the conversation.

- **Legal Knowledge Retrieval** – Actively searches and prioritizes the most relevant laws, regulations, legal precedents, and case judgments using Retrieval-Augmented Generation (RAG).
- **Contract Review & Risk Analysis** – Automatically analyzes contracts and legal documents to identify risky clauses, obligations, inconsistencies, and ambiguous language.
- **Compliance Verification** – Evaluates uploaded documents against applicable laws and regulations, identifies compliance gaps, and suggests corrective actions.
- **Case Research & Decision Support** – Retrieves relevant court cases, legal precedents, and regulatory references to provide users with accurate, context-aware legal insights and recommendations.
- **Conversational Legal Assistance** – Maintains context throughout multi-turn conversations, explains legal concepts in simple language, asks clarifying questions when necessary, and provides structured guidance while clearly stating that the information is for educational purposes only and not a substitute for professional legal advice.



# Novelty and Uniqueness

## Novelty and Uniqueness of the AI Legal Aid Multi-Agent System

The novelty and uniqueness of the **AI Legal Aid Multi-Agent System** lies in transforming complex legal information into an intelligent, interactive, and accessible platform for legal guidance and document analysis.

- **Unified Legal Retrieval** – Consolidates laws, regulations, contracts, and case judgments into one AI-powered platform using RAG.
- **Smart Document Analysis** – Reviews contracts and policies, highlighting key clauses, obligations, and potential legal risks automatically.
- **Agentic Legal Assistance** – Coordinates specialized AI agents for contract review, compliance checks, legal research, and user-specific Q&A.
- **Context-Aware Conversations** – Maintains conversation context, explains legal concepts in simple language, and provides personalized guidance.

## Git Hub Link

<https://github.com/umeirp3636-ux/AI-Legal-Aid-Multi-Agent-System>

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# Future Scope

1. **Advanced AI Integration** – Leveraging more powerful IBM Granite models for deeper legal reasoning, case analysis, and contextual legal assistance.
2. **Multimodal Support** – Expanding beyond text to support voice queries, document scanning, and image-based legal document analysis.
3. **Personalized Legal Guidance** – Tailoring responses and recommendations based on user profiles such as individuals, businesses, students, or legal professionals.
4. **Real-Time Legal Updates** – Integrating with live legal databases to provide the latest laws, regulations, court judgments, and policy changes.
5. **Government & e-Governance Integration** – Connecting with digital government services for document verification, legal forms, and citizen support.
6. **Global Legal Knowledge** – Supporting multiple jurisdictions and languages to provide legal assistance across different countries.
7. **Continuous Knowledge Expansion** – Automatically updating the knowledge base with new legislation, judicial precedents, and legal research publications.

# Certificates Earned:

In recognition of the commitment to achieve  
professional excellence



## Umeir Patel

Has successfully satisfied the requirements for:

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### Getting Started with Artificial Intelligence

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## Certificates Earned:



# Thank You!

Thank you for your time and interest.